
Exam A — 036026 Kinematics, Dynamics and Control of Robots

Spring Semester 2025

Exam information and instructions

- Duration: 14:00 - 17:00, June 23, 2025
- You are allowed to use only the lab computer at the workstation allocated to you. You are not allowed to use your own laptop or a tablet (like an iPad).
- The first 30 minutes of the exam is closed book, where you need to answer question 1 on Moodle, using the Safe Exam Browser. You are allowed to use a calculator to answer question 1. Question 1 consists of five short questions counting 3 marks each. You have only one opportunity to submit your answers; no repeated submissions are allowed. After 30 minutes (at 14:30) question 1 closes on Moodle and you will no longer have access to it. Make sure you submit your answers before it closes.
- After question 1 closes, the exam proceed to an open book part for the next 150 minutes, during which you are allowed to use paper notes and books. You are allowed to load data on the computer from a USB flash drive or external drive. You are also allowed to download files from your own space on Moodle. You may start with the open book part of the exam during the first 30 minutes of the exam, but you are not allowed to use your notes or books during the first 30 minutes.
- You are not allowed to use the internet except for uploading files or information to or from Moodle. You are not allowed to communicate with another person via email, WeChat or whatever other means possible. If you are caught using the internet — except Moodle — or communicating with a person other than the invigilators or the lecturer during the exam, this will be considered as academic dishonesty and disciplinary action against you will follow.
- You need to download various files that you need to use in answering questions 2, 3 and 4, from Moodle. All these files are in the zip file “Exam_A_files.zip”, which is available on Moodle under the “Exams” topic.
- The answers you present to the open book part of the exam should show a logical train of thought, starting from the theory that was covered in the reading assignments. If you use your computer to solve your problems, remember that the grader of your exam does not see your computer script and output. You must therefore supply enough written information in your exam script to allow the grader to assess your answer. If you solve equations symbolically or numerically in MATLAB or Python, these equations must be written in your exam script, with an appropriate and logical explanation of how they were derived.
- In final answers to questions, all fractions, products, sums, exponential expressions and natural constants like π should be evaluated to single real numbers, given accurately to four significant digits.
- For questions 2 and 3 you need to fill in variables or parameters that you have calculated in a feedback tool on Moodle. All the variables/parameters that you fill in in the feedback

tool, you must also clearly write in your exam script, so that it can be confirmed after the exam that the values entered on Moodle are actually the values that **you** have determined in the exam. If you fail to supply these parameters in your script, your answers on Moodle will be ignored and you will get zero for the corresponding part of the question. If you write the parameters in your script only, you will certainly lose marks.

- In all questions use $g = 9.81 \text{ m/s}^2$ where required.
- The notation used in this exam is the same as what is used in the course text book by Franklin and in the course presentation, unless indicated otherwise.
- The paper consists of four questions. Full marks are 100.